MIL-M-4673A(USAF)
15 April 1964
Superseding
MIL-M-4673 (USAF)
27 April 1953

MILITARY SPECIFICATION

MICROPHONE SIMULATOR SET AN/URM-14

1. SCOPE

1.1 This specification covers one type of microphone simulator set, designated AN/URM-14.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-J-641	Jacks, Telephone, General Specification for
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for

STANDARDS

Military

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6625

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3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The microphone simulator set shall consist of the following major components:

Description	See Requirement
Meter	3.6.1
Controls and hardware	3.6.2
Cables	3.6.3.

3.3 General requirements.- The requirements of MIL-T-21200, as specified for class 2 equipment, and MIL-STD-454 apply as requirements of this specification with the exceptions and additions specified herein. When the specification and standard conflict with this specification, this specification shall govern.

3.4 Design and construction.- The microphone simulator set shall be so designed and constructed that it will operate with an external audio oscillator connected to the input jack and a radio transmitter connected to the output jack. It may also have an ammeter connected to the microphone current jack. The circuit design shall be similar to that depicted on figure 1.

3.4.1 General configuration.- The microphone simulator shall be housed in a combination case and shall have a configuration as shown on figure 2. The meter and all hardware shall be mounted in the approximate locations as indicated.

3.4.2 Reliability.- The microphone simulator set shall be designed to give a low probability of failure in normal use and to meet the tests specified herein without failure.

3.4.3 Maintainability.- The equipment shall be designed and constructed to provide for rapid and simple identification of failure and accessibility to components.

3.4.4 Primary power.- There shall be no primary power required by the microphone simulator set.

3.4.5 Electromagnetic interference.- Interference control requirements of MIL-T-21200 shall not apply.

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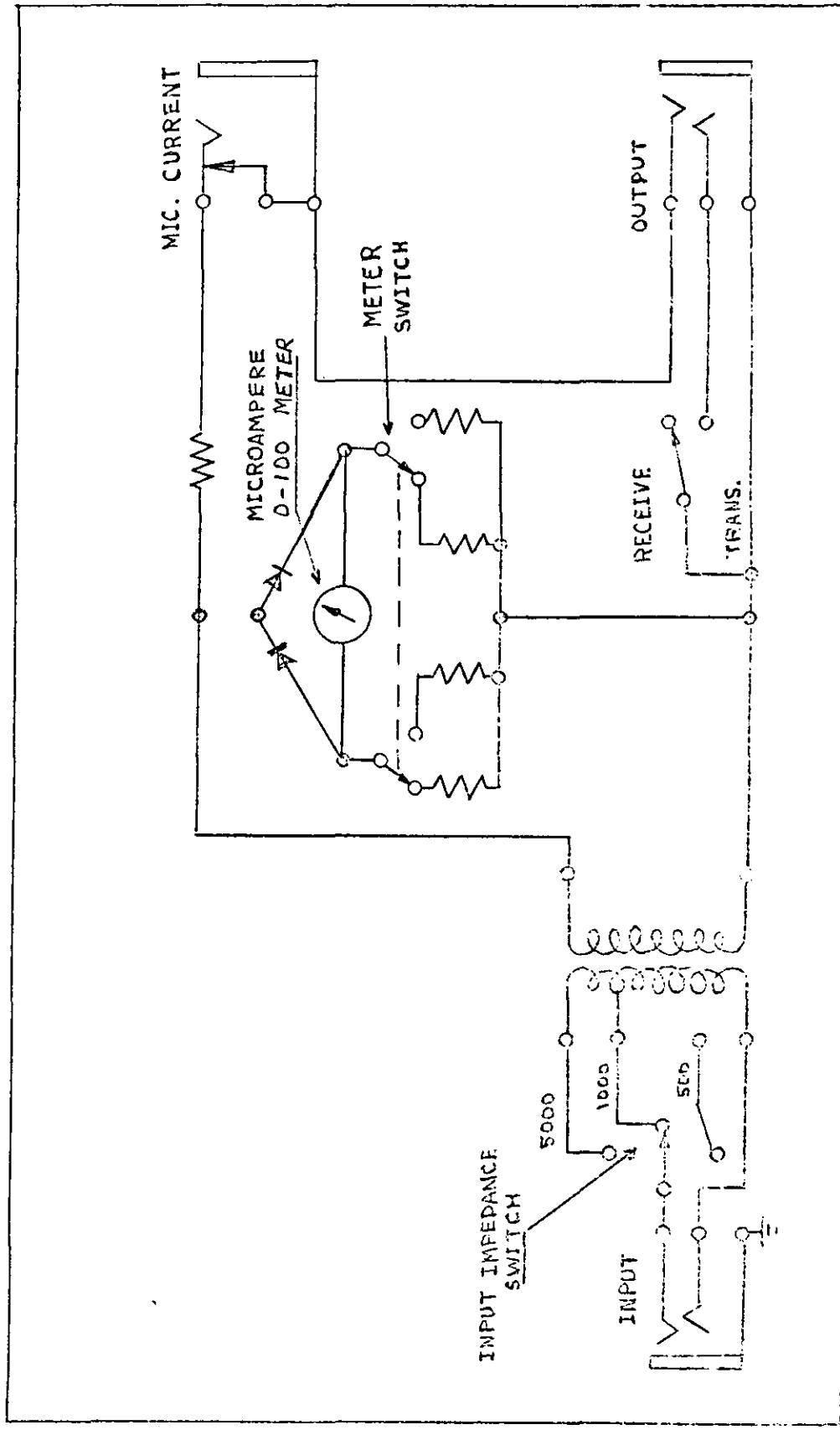
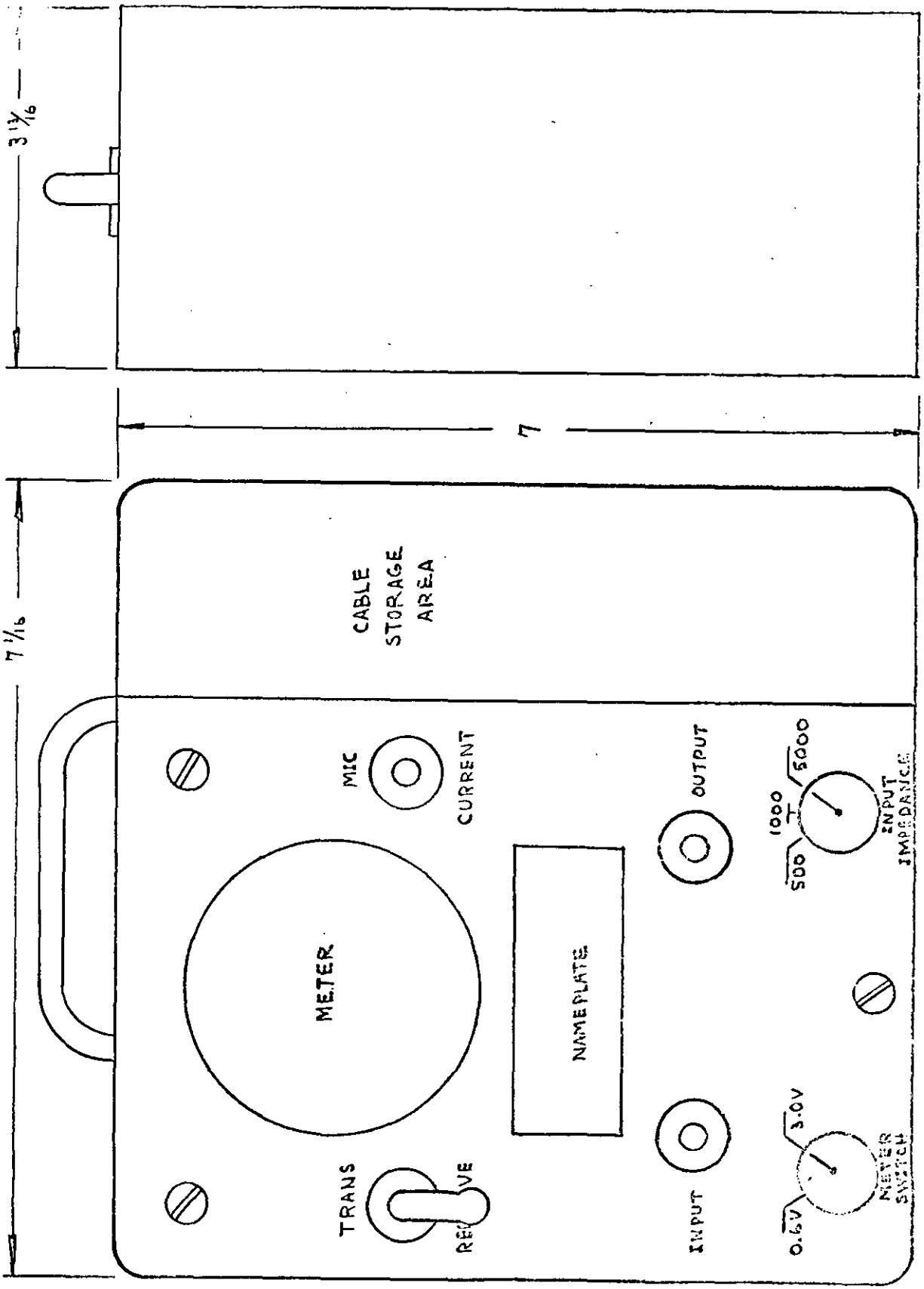


FIGURE 1. Typical Circuit

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DIMENSIONS IN INCHES

FIGURE 2. Microphone Simulator Set Configuration

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3.5 Performance

3.5.1 Environmental conditions.- The microphone simulator set shall be capable of meeting the environmental conditions of MIL-T-21200 except as modified herein:

a. Temperatures

(1) Operating - continuous or intermittent: -40° to +55° C

(2) Nonoperating: -62° to +71° C

b. Humidity: 95 percent

c. Requirements for sand and dust, salt-sea atmosphere, and explosive conditions shall not apply.

3.5.2 Input characteristics

3.5.2.1 Frequency.- The microphone simulator set shall operate with input frequencies of 300 to 4,000 cps.

3.5.2.2 Impedance.- The input impedance shall be 500, 1,000 or 5,000 ohms as indicated by the position of the impedance switch. The impedance shall be provided by means of an input transformer with appropriate taps.

3.5.3 Output characteristics

3.5.3.1 Output voltage.- The actual open circuit output of voltage as recorded by an external vacuum tube voltmeter (VTVM) shall be within ±5 percent of the voltage reading of the microphone simulator voltmeter.

3.5.3.2 Output impedance.- The output impedance shall be approximately 82 ohms.

3.6 Details of components

3.6.1 Meter.- A 2-1/2-inch, 0- to 100-microampere ammeter with two voltage scales shall be provided. One scale shall be a 0.6-V scale with markings from 0 to 0.6V full scale in 0.1-V increments. The other scale shall be a 3-V scale with markings from 0 to 3V in 0.5-V increments. The meter accuracy shall be at least ±3 percent of full scale.

3.6.2 Controls and hardware

3.6.2.1 Transmit/receive toggle switch.- A toggle switch shall be provided that will place the transmitter being tested into the transmit mode if the transmitter has the provision for a press-to-talk carbon microphone.

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3.6.2.2 Meter switch.- The meter switch shall be a two-position, one-section, two-circuit switch which sets the range of the voltmeter to either 0.6V or 3V full scale.

3.6.2.3 Input impedance switch.- A three-position, one-section, one-circuit impedance switch shall actuate the input impedance to 500, 1,000, or 5,000 ohms as may be desired.

3.6.2.4 Input and output jacks.- One each input and output jack conforming to MIL-J-641, type JJ-033, shall be provided.

3.6.2.5 Microphone current jack.- A microphone current jack conforming to MIL-J-641, type JJ-089, shall be provided.

3.6.3 Special purpose cables.- Two special purpose cables, each 5-feet in length, shall be provided and shall plug into the jacks of the microphone simulator as specified herein:

a. Special purpose cable assembly CX-1296/U.- This cable shall be a 3-conductor, stranded copper wire of suitable size and covered by a rubber sheath. The cable shall be terminated with a PJ-068 plug, or equal, on each end and may be used either as an input or output cable when using the radio frequency generator.

b. Special purpose cable assembly CX-1299/U.- This cable shall be a 2-conductor, stranded copper wire of suitable size and covered by a rubber sheath. The cable shall have a molded plug (General Radio type 274-71SF, or equal) on one end to permit pickup from audio sources using fittings which will accommodate this type of plug. The other end shall terminate with a PJ-068 plug, or equal, which will plug into the input jack of the microphone simulator.

3.7 Weight.- The weight of the microphone simulator set shall not exceed 5 pounds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection and testing.- The inspection and testing of the microphone simulator set shall be in accordance with the applicable tests of MIL-T-21200 with the exceptions and additions specified herein.

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4.3 Classification of tests.- The inspection and testing of the microphone simulator set shall be classified as follows:

a. Preproduction testing. . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.4 Preproduction testing

4.4.1 Test samples.- The preproduction test samples shall consist of three microphone simulator sets.

4.4.2 Test report.- After the preproduction tests are completed, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity.

4.4.3 Preproduction tests.- The preproduction tests shall consist of all the tests described under 4.6. The preproduction samples shall be tested as follows:

Sample No. 1	Sample No. 2	Sample No. 3
Examination of product	Examination of product	Examination of product
Input impedance	Input impedance	Input impedance
Output impedance	Output impedance	Output impedance
Output voltage	Output voltage	Output voltage
Low pressure	Vibration	Humidity
High temperature	Shock	Fungus
Low temperature	Drop test (on case only)	Watertight test (on case only)

4.5 Acceptance tests.- The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests.- Each microphone simulator set shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Input impedance

c. Output impedance

d. Output voltage.

4.6 Test

4.6.1 Examination of product.- Each microphone simulator set shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and weight. In addition, all moving parts shall be examined to insure that they operate freely without sticking or binding.

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4.6.2 Input impedance.- The input impedance of 500, 1,000, and 5,000 ohms shall be measured while an 82-ohm resistive load is connected across the output terminals of the microphone simulator set. The measurement shall be made with an input frequency of 1,000 cps and with a 0.6-V reading on the microphone simulator voltmeter. The measured value shall be within ±10 percent of the nominal value set on the input impedance switch.

4.6.3 Output impedance.- The output impedance of the microphone simulator set shall be measured and shall be 82 ±10 ohms at a frequency of 1,000 cps.

4.6.4 Output voltage.- A CX-1299/U cable shall be connected between the input jack and an audio oscillator (Hewlett Packard Model 200 CD, or equivalent). An input impedance match shall be provided at each input impedance setting. An a-c VTVM (Hewlett Packard Model 400C, or equivalent) shall also be connected across the output terminals. Twenty-four output voltage checks shall be performed with settings and readings as specified in table I.

4.6.5 Environmental tests.- The microphone simulator set shall be subjected to the following tests conducted in accordance with the specified methods of MIL-STD-810:

a. Low pressure - Method 500, procedure I

b. Temperatures

(1) High temperature - method 501

(2) Low temperature - method 502

c. Humidity - method 507, procedure I, except the number of humidity cycles shall be five

d. Shock - method 516, procedure I.

4.6.5.1 Fungus.- Certification by the contractor that the applicable portions of MIL-T-21200 have been complied with will negate the need for fungus testing. If noninert materials are used, the fungus test shall be in accordance with method 508, procedure II.

4.6.5.2 Vibration test.- The microphone simulator set shall be subjected to a vibration test in accordance with method 514, class 7 equipment, except that the test curve used shall be a step function as defined by the "frequency versus double amplitude" vibration requirements of MIL-T-21200.

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Table I

Output Voltage Checkpoints

Test	Frequency of Audio Oscillator (CPS)	AN/URM-14 Input Impedance Setting (Ohms)	AN/URM-14 Meter Scale Reading (Volts)		VTVM Reading (Volts)
1	300	500	0.6	0.1	0.1 ±0.03
2	300	1,000	0.6	0.1	0.1 ±0.03
3	300	5,000	0.6	0.1	0.1 ±0.03
4	4,000	500	0.6	0.1	0.1 ±0.03
5	4,000	1,000	0.6	0.1	0.1 ±0.03
6	4,000	5,000	0.6	0.1	0.1 ±0.03
7	300	500	0.6	0.6	0.6 ±0.03
8	300	1,000	0.6	0.6	0.6 ±0.03
9	300	5,000	0.6	0.6	0.6 ±0.03
10	4,000	500	0.6	0.6	0.6 ±0.03
11	4,000	1,000	0.6	0.6	0.6 ±0.03
12	4,000	5,000	0.6	0.6	0.6 ±0.03
13	300	500	3	0.5	0.5 ±0.15
14	300	1,000	3	0.5	0.5 ±0.15
15	300	5,000	3	0.5	0.5 ±0.15
16	4,000	500	3	0.5	0.5 ±0.15
17	4,000	1,000	3	0.5	0.5 ±0.15
18	4,000	5,000	3	0.5	0.5 ±0.15
19	300	500	3	3	3 ±0.15
20	300	1,000	3	3	3 ±0.15
21	300	5,000	3	3	3 ±0.15
22	4,000	500	3	3	3 ±0.15
23	4,000	1,000	3	3	3 ±0.15
24	4,000	5,000	3	3	3 ±0.15

4.6.6 Reliability test.- Design and construction in accordance with the requirements and completion of the tests specified herein without failure shall be considered demonstration of satisfactory reliability.

4.6.7 Maintainability verification.- Examination and inspection assuring compliance with the design, construction, and maintenance provisions specified herein shall be considered verification of satisfactory maintainability.

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5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with the instructions of the procuring activity.

6. NOTES

6.1 Intended use.- The AN/URM-14 microphone simulator set is designed to simulate the electrical characteristics of Microphone T-17 and Microphone ANB-M-C1 and is intended for use in testing radio transmitters.

6.2 Ordering data.- Procurement documents should specify the title, number, and date of this specification.

6.3 Technical guidance.- The following documents are listed as guides for complying with the requirements specified herein:

a. Technical Order 33A1-12-4-2 - Technical Manual Service Instructions, Microphone Simulator Set AN/URM-14

b. Technical Order 33A1-12-4-24 - Illustrated Parts Breakdown, Microphone Simulator Set AN/URM-14.

Custodian:
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SPECIFICATION ANALYSIS SHEET	Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use of the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD Form 1426

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